Technische Universiteit Eindhoven University of Technology

Student name:

Student number:

# Examination cover sheet

**(to be completed by the examiner)**

Course name: Particle-based simulations Course code: 6EMA02

Date: 1 November 2017

Start time: 13:30 End time: 16:30

Number of pages: 4

Number of questions: 4

Maximum number of points/distribution of points over questions: 25-25-25-25

Method of determining final grade: Grade written exam: total points divided by 10. The grade for the written exam has to be at least 5.0.

The final grade for the course is computed as: 50% assignment + 50% written exam.

Exam inspection: E.A.J.F. Peters

Other remarks: You are allowed to answer questions in English or Dutch. Sometimes you are asked to use an earlier result in a subsequent question. If you do not have this result assume a value and explicitly state the assumption.

## Instructions for students and invigilators

**Permitted examination aids (to be supplied by students):**

This is a closed book exam. So no material other than those stated below are allowed

□ Calculator

□ Dictionar(y)(ies). If yes, please specify:any bilangual dictionary

**Important:**

- examinees are only permitted to visit the toilets under supervision
- it is not permitted to leave the examination room within 15 minutes of the start and within the final 15 minutes of the examination, unless stated otherwise
- examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- the house rules must be observed during the examination
- the instructions of examiners and invigilators must be followed
- no pencil cases are permitted on desks
- examinees are not permitted to share examination aids or lend them to each other

**During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:**

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, etc.
- using a clicker that does not belong to you
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- visiting the toilet (or going outside) without permission or supervision

## Section 1. General aspects of particle-based simulations (25 points, 5 each)

a) Describe the similarities and differences between, microscopic, mesoscopic and macroscopic particle-based simulation methods.

b) In a simulation particles 1 and 2 have a bonded interaction described by the stretch potential:

$$\varphi(r_{12}) = \frac{k}{2}\,(r_{12} - r_0)^2, \text{where } r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$$

Derive the expression for all three components $(x, y, z)$ of the corresponding force experienced by particle 2.

c) Which of the following forces that act on a particle located at position $(x, y, z)$ are conservative? Give a motivation for your answer for each of the 5 forces.

1) $\mathbf{F} = \begin{pmatrix} -x/r^3 \\ -y/r^3 \\ -z/r^3 \end{pmatrix}$, with $r = \sqrt{x^2 + y^2 + z^2}$

2) $\mathbf{F} = \begin{pmatrix} -v_x/r^3 \\ -v_y/r^3 \\ -v_z/r^3 \end{pmatrix}$, with $r = \sqrt{x^2 + y^2 + z^2}$

3) $\mathbf{F} = \begin{pmatrix} -x/r^3 \\ -y/r^3 \\ 0 \end{pmatrix}$, with $r = \sqrt{x^2 + y^2}$

4) $\mathbf{F} = \begin{pmatrix} y/r^3 \\ -x/r^3 \\ 0 \end{pmatrix}$, with $r = \sqrt{x^2 + y^2}$

5) $\mathbf{F} = \begin{pmatrix} y \\ x \\ 0 \end{pmatrix}$

d) In a particle-based simulation with periodic boundary conditions of $N$ particles in volume $V$, a neighbor list is used with a force cutoff distance $r_{cut}$ and a shell distance $r_{shell}$. Estimate how many distance computations are typically performed inside the non-bonded pair-force computation routine. You can assume that the particles are distributed uniformly in the simulation cell.

e) Write down a Verlet or leap-frog time integration scheme in pseudo or C code.

## Section 2. Microscopic Simulations (25 points, 5 each)

You are asked to perform a molecular simulation of the noble gas Xenon. The interaction between Xenon molecules can, in good approximation, be described by means of Lennard-Jones pair interactions:

$$\varphi(r) = 4\varepsilon \left( \left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right),$$

with parameters $\varepsilon/k_B = 234.7$ K and $\sigma = 4.009$ Å (and $M = 131.3$ g/mol).

a) According to the kinetic theory of dilute gases the self-diffusion coefficient is well estimated by

$$D = \frac{3}{8}\sqrt{\frac{(RT)^3}{\pi\, M}}\,\frac{1}{p\, N_{AV}\, \sigma^2\, \Omega(T^*)}, \text{with } T^* = \frac{k_B T}{\varepsilon} \text{ and}$$

$$\Omega(T^*) = \frac{1.06036}{T^{*0.15610}} + \frac{0.19300}{\exp(0.47635\, T^*)} + \frac{1.03587}{\exp(1.52996\, T^*)} + \frac{1.76474}{\exp(3.89411\, T^*)}.$$

Compute the self-diffusion coefficient at 1 bar and 293 K in SI units.

b) Give the value of the (1 bar) pressure and the self-diffusion coefficient in LJ units.

c) The diffusion coefficient can be computed from an velocity autocorrelation function by

$$D = \frac{1}{3N}\sum_{i=1}^{N}\int_0^\infty \langle \mathbf{v}_i(0)\cdot\mathbf{v}_i(t)\rangle\, dt$$

This velocity autocorrelation function is found to be well approximated by

$$\langle \mathbf{v}_i(0)\cdot\mathbf{v}_i(t)\rangle = \alpha \exp\left(-\frac{t}{\tau}\right)$$

What are the expected values for $\alpha$ and $\tau$ in LJ units?

d) You would like to compute the diffusion coefficient at 1 bar and 293 K using a NPT (constant temperature and pressure) simulation using 1000 particles. Explain in sufficient detail how you can control the temperature and pressure in a molecular simulation.

e) Give an estimate of the number of time steps you need to determine the diffusion coefficient within 1% accuracy.

## Section 3. Mesoscopic Simulations (25 points, 5 each)

You are asked to model the sedimentation of colloidal particles with a radius of $R_c = 1$ μm in water. The colloidal particles have a relative mass density of 1.2 (with respect to water). You perform a stochastic rotation dynamics (SRD) simulation with the following SRD settings:

| SRD quantities | symbol | value |
|---|---|---|
| number density of SRD particles | $\gamma$ | $10\ a_0^{-3}$ |
| collision interval | $\delta t_c$ | $0.1\ a_0\ m^{1/2}\ (k_B T)^{-1/2}$ |
| rotation angle | $\alpha$ | $\pi/2$ |
| dynamic viscosity | $\mu$ | $5.43\ a_0^{-2}\ m^{1/2}\ (k_B T)^{1/2}$ |
| sound velocity | $v_s$ | $\sqrt{5/3}\, m^{-1/2}\ (k_B T)^{1/2}$ |
| other constants | | |
| Boltzmann constant | $k_B$ | $1.38064852 \times 10^{-23}\ \mathrm{m^2\ kg\ s^{-2} K^{-1}}$ |

Here $a_0$ is the SRD cell size and $m$ the mass of an SRD particle. Note that $\gamma$, when expressed in SRD units, can also be interpreted as the average number of SRD particles per SRD cell: in this case there are on average 10 SRD particles per cell.

a) In a first (naïve) approach you try to pick values of $a_0$ and $m$ such that you simulate water at $T = 292$ K, $\rho_w = 998\ \mathrm{kg/m^3}$ and $\mu_w = 1.00 \cdot 10^{-3}$ Pa s. What are the corresponding values of $a_0$ and $m$ in SI units? What is the radius of the colloidal particle expressed in terms of this $a_0$? Why is this approach unfeasible (i.e. practically impossible)?

b) Explain the proper methodology to set the parameters in an SRD simulation. According to this methodology, what is a good value for $a_0$ in SI units?

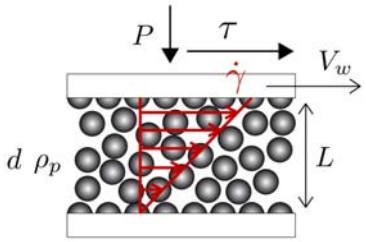
c) Consider sedimentation due to gravity. The drag-coefficient on a single particle is given by the Stokes relation: $\zeta = 6\pi\mu_w R_c$. The self-diffusion coefficient of a colloidal particle follows from the Stokes-Einstein relation: $D = k_B T/\zeta$. Determine the final settling velocity of a single colloidal particle from a force balance between the drag force and other forces. What is the Peclèt number (convection / diffusion) for a single sedimenting colloidal particle? Can this sedimentation be faithfully modeled using SRD?

d) To relate the simulation results to experimental results: give the appropriate value of the gravitational acceleration used in the SRD code in SRD units and the value of $m$ in SI units.

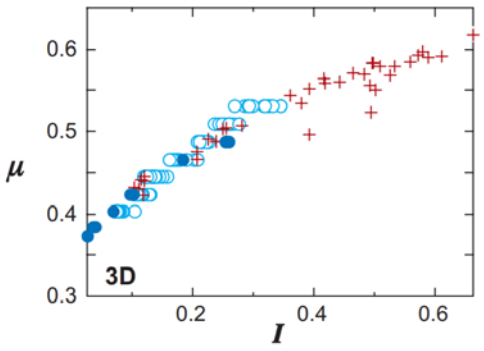
e) You would like to model a no-slip boundary condition at the surface of the colloidal particle. How would you do that?

## Section 4. Macroscopic Simulations (25 points, 5 each)

You want to simulate a granular material made up of mono-disperse spheres in shear flow at a supplied normal stress (normal force per unit area), $P$, by moving a rough upper plate as depicted here:




It is found that the shear stress (i.e. the tangential force acting on the top wall per unit area) obeys the following relation:




$$\tau = P\,\mu(I), \text{with } I = \frac{\dot{\gamma}\,d}{\sqrt{P/\rho_p}}.$$

$\tau$: shear stress
$\dot{\gamma} = V_w\,/L$: shear rate
$d$: sphere diameter
$P$: normal stress
$\rho_p$: solids mass density

The graph is taken from Forterre and Pouliquen, Annu. Rev. Fluid Mech. 2008. 40:1–24. It contains a compiled data set of experimental and simulation results. Surprisingly the shear stress (divided by $P$) only depends on the dimensionless number, $I$, which is called the inertial number. This relation is found to be the same, irrespective of the values used for friction coefficients and restitution coefficients (as long as they are dissipative). The aim of your simulation is to reproduce this result.

a) To model the particle-particle interactions you use the soft-sphere model. Describe the main features of this model.
b) Describe the boundary conditions you would apply in your simulations.
c) How would you impose the normal stress $P$ in the simulation?
d) How would you compute $\tau$ from the simulation?
e) The ratio $\mu = \tau/P$ is commonly called the 'macroscopic friction coefficient'. Explain why this is a suitable name for $\mu$.